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Event Detection in Water Distribution Systems from Multivariate Water Quality Time Series

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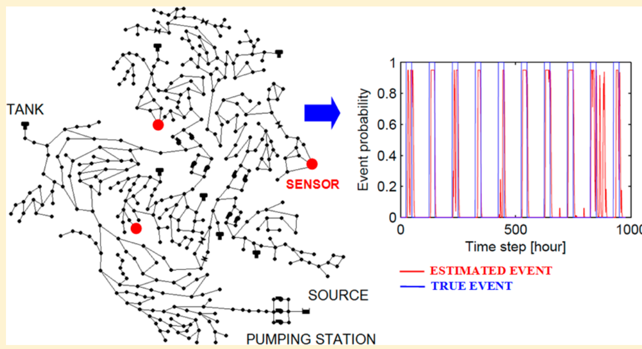
Supporting Information

ABSTRACT: In this study, a general framework integrating a data-driven estimation model with sequential probability updating is suggested for detecting quality faults in water distribution systems from multivariate water quality time series. The method utilizes artificial neural networks (ANNs) for studying the interplay between multivariate water quality parameters and detecting possible outliers. The analysis is followed by updating the probability of an event, initially assumed rare, by recursively applying Bayes' rule. The model is assessed through correlation coefficient (R^2), mean squared error (MSE), confusion matrices, receiver operating characteristic (ROC) curves, and true and false positive rates (TPR and FPR). The product of the suggested methodology consists of alarms indicating a possible contamination event based on single and multiple water quality parameters. The methodology was developed and tested on real data attained from a water utility.



INTRODUCTION

Securing critical infrastructure is vital for ensuring society's well being. Water distribution systems (WDS) are inherently vulnerable as they comprise of numerous exposed elements, which can be exploited for malicious actions.[1] Physically securing each of these apparatuses is not feasible, thus other methods ensuring the delivery of sufficient (quantity) and adequate (quality) drinking water need to be developed. In recent years, there have been fast developments in wireless/wired sensor networks (WSN) for various applications such as environmental monitoring, infrastructure security, and water distribution systems monitoring. Application of WSN for water distribution is being extensively studied employing different sensors capable of continuously collecting and transmitting hydraulic and water quality measurements at fine temporal resolution facilitating an accurate constantly updating representation of the conditions in the distribution system.[2,3] This allows for an improved online decision support system for analyzing, modeling, and controlling water supply systems.

The application of WSN in water supply systems for securing water quality principally focuses on: (1) selecting water quality parameters that are good indicators of contamination, (2) deciding on the number and locations of sensors in a water distribution system, and (3) performing temporal data analyses for identification of possible quality faults.

To address the first topic, the Environmental Protection Agency (EPA) and online water quality sensor manufacturers (Hach, Rosemount, Clarion, ManTech, Analytical Technology) are conducting research, under the Environmental Technology Verification (ETV) program established in 1995, and have provided information about the potential contaminants that produce detectable changes in online measured water quality parameters.[4] Thirty-three contaminants (pesticides, insecticides, metals, bacteria, etc.) that are of particular concern to intentional water contamination were tested and analyzed by the EPA. Figure S1 of the Supporting Information shows three selected contaminants that triggered multiparameter probes during ETV tests. It was established that information from online water quality sensors already deployed in water distribution systems, which are reactive to many contaminants, can provide an early indication of possible pollution, as opposed to intermittent grab sampling and analysis of a number of specific contaminants, which may be limited in its ability to alert about a potential contamination.[5,6]

The problem of selecting and locating a number of monitoring stations has been widely studied including deterministic and stochastic optimization techniques and graph-theory algorithms optimizing one or more objectives as detection likelihood, expected contaminated water volume and affected population, and design cost.[7−12]

Several related works are conducted on the third topic aiming at finding quality faults to enhance system security. The *CANARY* quality event detection software has been developed at Sandia National Laboratories in collaboration with the EPA's National Homeland Security Research Center to provide both

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off-line and real-time analysis tools for the monitored data to detect anomalous changes from the baseline and indicate possible contamination.[13−16] The HACH GuardianBlue Early Warning System analyzes monitored data, finds significant deviations from baseline, matches the event to previous patterns through events fingerprint, and alarms about possible events.[17] Other software packages are being developed for monitoring station management, data validation, and event detection.[18]

This work focuses on the third problem of detecting potential contaminants in the water distribution system. A number of models have been developed to understand and model multicomponents in a WDS, such as the chlorine decay, reformation of microbial contaminants, and substrate consumption; however, the application and calibration of these models is a difficult task due to the number of parameters involved and the information required.[19−21] The proposed methodology utilizes artificial neural networks (ANNs) to estimate the relationships between water quality parameters in a WDS. ANNs have attracted an undisputed amount of interest over the past two decades and have been successfully applied for prediction and modeling phenomenon in various fields. A few studies applying ANNs in water analysis and water quality assessment have focused on parameters modeling and prediction; including prediction of trihalomethanes formation, prediction of residual chlorine, and prediction of the temporal variation of residual chlorine, substrate, and biomass concentrations.[22−28] Although the functional form of the reactions of some of the chemicals is known, such as chlorine decay with time, the relationships between water quality parameters are unknown. Because the development of ANN models does not require a priori knowledge of the physical and chemical laws governing the water parameters, it makes them an attractive tool for modeling multivariate reactions in water supply systems.

The proposed method suggests detecting potential contamination events by combining an ANN model for temporary analysis of multivariate water quality time series with Bayesian sequential analysis for estimating the probability of an event. A similar approach has been previously suggested for detecting hydraulic faults in the system.[29,30]

METHODOLOGY

The proposed scheme for event discovery relies on multivariate time series data collected by a supervisory control and data acquisition (SCADA) system sensing hydraulic and water quality data gathered from a WDS. Historical data is used for the data-driven model training, error threshold setting, and model assessment. New incoming observations are used to discover possible quality faults in real time. The main steps of the proposed algorithm are described in subsequent sections and depicted in Figure 1.

Artificial Neural Networks. In the first stage, ANNs are created and trained for processing characteristics and modeling the relationships between multivariate water quality parameters in a water distribution system. One of the most common network architectures are multi-layer perceptrons (MLPs using the back-propagation algorithm for training for minimizing the prediction error made by the network.[31,32] MLPs are generally represented by an input layer, a hidden layer, and an output layer and correlate input variables to output variables through nonlinear, weighted, parametrized functions. The design of the ANN depends on: (1) hardware − number of hidden layers and number of neurons in hidden layers;[33] (2) software − transfer (activation and output) functions and performance function.[34] Data is passed from the input layer with p inputs $x = (x_1,...,x_p)$ to the hidden layer having m neurons. Each node in the hidden layer receives the summed weighted outputs of the preceding layer, which are then passed through an activation function to produce the hidden node output. The output layer with K targets $y = (y_1,...y_k)$ again, receives the summed weighted outputs of the preceding layer, which are passed through an output function, and the final output is estimated by the function f_k $(x;w)$. The mathematical model and graphical representation of an MLP are shown in eq 1 and in Figure S2 of the Supporting Information, respectively.

What's the structure of the ANN that I have implemented?

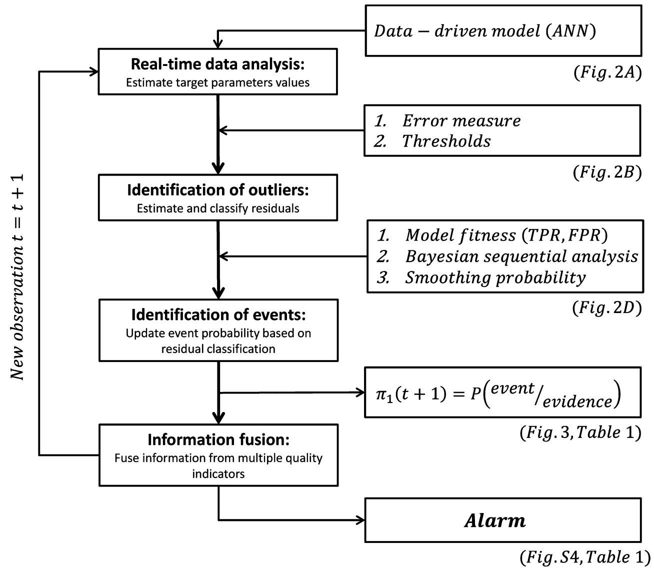

Figure 1. Main stages of the event detection methodology.

$$f_k(x;\ w) = \varphi_0[w_0 + \sum_j w_{jk}\varphi(w_{0j} + \sum_i w_{ij}x_i)] \quad (1)$$

where w_{jk}, w_{ij} are weights, w_0, w_{0j} are biases, φ is the activation function, φ_0 is the output function, and f_k $(x;w)$ is used to estimate the target value y.

In this work, ANNs are utilized to correlate water quality parameters through nonlinear, weighted, parametrized functions during normal operation. The ANNs are suitable for this purpose because their development does not require a priori knowledge of the physical and chemical laws governing the water quality parameters. An ANN model is constructed for each target quality parameter with the input vector containing measured time series of all predictive parameters and lagged target parameter, as formulated in eq 2. The network is trained by repeatedly presenting the input data, comparing its output to the desired output, and calculating the performance function − mean squared error (MSE). The error is back-propagated to the network and is used to adjust the weights so that the error decreases with each iteration.

$$\hat{x}_i(t) = f(x_1(t), ..., x_{i-1}(t), x_i(t-1), x_{i+1}(t), ..., x_n(t)) \quad (2)$$

Where $x_i(t)$ and $\hat{x}_i(t)$ are the measured and estimated water parameters at time t respectively, and $f(\cdot)$ is defined by the ANN, as in eq 1. The fit of the model is evaluated through mean, standard deviation (STD), and correlation (R^2) between the measured and estimated water quality parameters.

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Corresponds to the anomaly score

Residual Estimation and Classification. The next step is to estimate and classify residuals. Residuals are estimated as the difference between the measured and estimated parameters' values, as shown in eq 3, represented as time series.

$$ER_i(t) = x_i(t) - f_i(\cdot) = x_i(t) - \hat{x}_i(t) \tag{3}$$

where $ER_i(t)$ is the estimated residual for parameter i at time step t.

For each ANN model, the estimated residuals are bounded such that the majority of the errors rest within the upper and lower limits. Observations exceeding the threshold are considered to be outliers. Intuition: The perception is that during normal operation conditions water quality parameters exhibit some associative behavior and correlation, whereas during events the interplay between some or all of the water quality parameters will change compared to normal conditions and the model will result in larger errors.

Same explanation in the final report

Sequential Bayesian Updating. In this stage, the probability of an event is updated using sequential Bayesian analysis for each new observation. In sequential analysis, the number of observations is not known in advance; instead, observations come in sequence, and a decision needs to be made about the current state. Each time, three possible actions can be made: nonevent, event, or take additional observation. Initially, the probability of an event is assumed rare and with each new observation the posterior probability of an event is sequentially updated using Bayes' rule, as shown in eq 4.

Given $P(\theta = \theta_1) = \pi_1$ and $\pi_1(t) = P(\theta = \theta_1 | y_{1:t})$ the posterior probability is:

$$\pi_1(t+1) = \frac{P(y_{t+1}|\theta_1)\pi_1(t)}{P(y_{t+1}|\theta_1)\pi_1(t) + P(y_{t+1}|\theta_0)(1 - \pi_1(t))} \tag{4}$$

where θ_1 = event, θ_0 = nonevent, and $y \in$ {outlier, normal}.

The outcome of the probability projects on the decision considering a possible event. Initial prior probability is set to $\pi_1(0) = \pi_0$, $P(y_{t+1}|\theta_1)$ and $P(y_{t+1}|\theta_0)$ are true positive rates (TPR) and false positive rates (FPR) respectively, which are metrics of models' performance during events.

The performance of the ANN models is measured through confusion matrix during the training of the proposed algorithm. The confusion matrix represents the model's classification of all observations to one of four classes: True positive − the residual was classified as an outlier during an actual event; false positive − the residual was classified as an outlier during routine operation; true negative − the residual was classified as plausible model error during routine operation; and false negative − the residual was classified as plausible model error during an actual event. Sensitivity and specificity are commonly used metrics that can be derived from confusion matrix as shown in eq 5:

$$\begin{aligned} \mathrm{TPR} &= \frac{TP}{TP + FN} = \text{sensitivity} \\ \mathrm{FPR} &= \frac{FP}{FP + TN} = 1 - \text{specificity} \end{aligned} \tag{5}$$

TPR and FPR are also used to construct receiver operating characteristic (ROC) curves, which visually depict the same information as the confusion matrix demonstrating the fundamental performance trade-off between TPR and FPR in a much more intuitive fashion.[35]

The probability of an event is updated using sequential Bayesian rule (eq 4), depending on the new observation being classified as an outlier or not, and on the TPR and FPR (eq 5). The posterior probability is updated independently for each water quality parameter and designates the likelihood of an event based solely on the target parameter. An event is declared when the probability exceeds some threshold value,

To avoid quick convergence, the updated probability is smoothed, as in eq 6 using a smoothing parameter $0.3 \le \alpha \le 0.9$.[36]

$$\pi_1(t+1) = \alpha\pi_1(t+1) + (1-\alpha)\pi_1(t) \tag{6}$$

Fuse Decision. At each time step, univariate event probabilities are fused to give a unified multivariate event probability reflecting the likelihood of an event based on all parameters. Each water quality parameter is assigned a weight reflecting its influence on the synchronized decision, for example, uniform given no prior information, proportional to the area of its ROC curve, or based on expert's opinion. Again, a critical probability, P_{Thres}, is defined to declare an event.

APPLICATIONS AND RESULTS

Data Preparation. *1. Data Acquisition and Partitioning.* The methodology is tested on real data collected by a utility in the United States and is available from *CANARY*.[15] The attained data was collected by the SCADA system sensing hydraulic and water quality data gathered from a WDS. The data contains online multivariate water quality measurements taken every 5 min during roughly 4 months (~35 000 time steps) during normal operating conditions and includes the following water quality parameters: Total chlorine, electrical conductivity (EC), pH, temperature, total organic carbon (TOC), and turbidity. The data was divided into two subsets, 67% for training and 33% for testing. The training subset is exploited to create, train, and evaluate the data-driven model through confusion matrix. The testing subset is exploited to imitate a real-time operation and to test the power of the suggested method to identify contamination events.

2. Event Simulation. Contamination events in water distribution systems heavily depend on the environmental factors making the events hard to be prespecified. To cope with this difficulty, contamination events were simulated and superimposed on routine patterns, characterized by the magnitude, direction, and duration of deviation.[37] The shape of the events was assumed to be Gaussian, with the duration of 8 h for all events, and with randomly sampled deviation for each water quality parameter from routine patterns in the range between 0.5 and 2.5. Figure S3 of the Supporting Information shows a partial time series plot of all water quality parameters during routine operation and randomly simulated events.

We only have flow and chlorine as parameters and we probably just need one network to predict only chlorine

Model Preparation. *1. Data Driven Model.* Six ANN models were created and trained, one for each quality parameter $x_i(t)$ $i = 1,...,6$ with water parameters {total chlorine, EC, pH, temperature, TOC, turbidity} respectively to estimate the target water quality parameters and the relationships between them. The development of an ANN model involves defining the size of the ANN, its inputs, and a training algorithm and testing the developed model with unseen data. In this work, a feed-forward back-propagation network in one hidden layer and one output layer was constructed. The network was trained with tan-sigmoid transfer function in the hidden layer and linear transfer function in the output layer to approximate the interplay between the parameters. This

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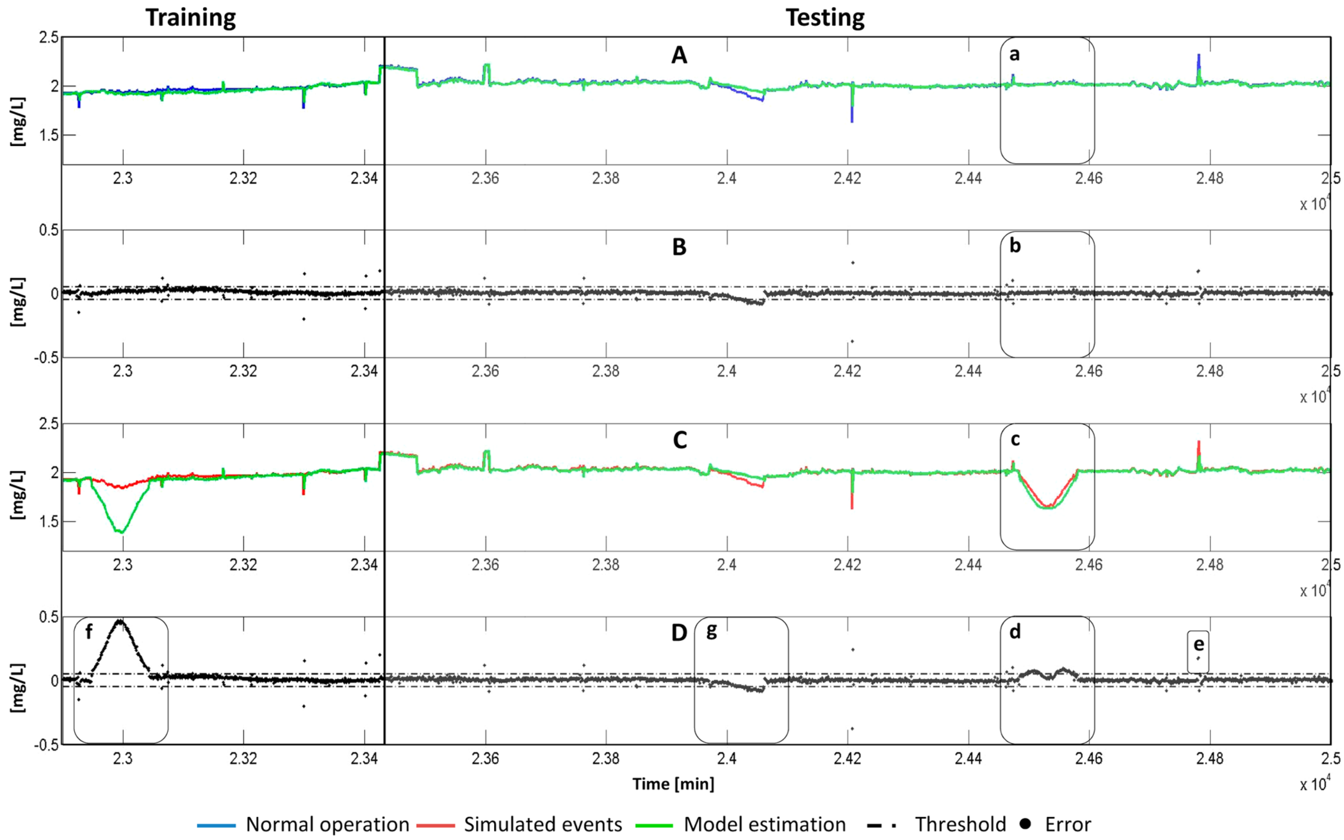

Figure 2. Total chlorine partial training and testing sets. (A) Measured (blue) and estimated (green) values, (B) error between measured and estimated values, (C) simulated events (red) and estimated (green) values, (D) error between simulated events and estimated values; **(a–d)** 24 484–24 579 [time steps] – **(a)** normal operation, **(b)** error, **(c)** contamination event, **(d)** multiple outliers (true positive), **(e)** an outlier, **(f)** multiple outliers [22 948–23 427] (true positive), **(g)** multiple outliers [24 035–24 065] (false positive).

structure is useful for function approximation or regression problems. One model was constructed for each target quality parameter with the input vector containing measured time series of all predictive parameters at the same time and lagged target parameter according to eq 2, that is a total of six inputs for each time step. For example, the model for estimating total chlorine takes the following inputs:

$$\hat{x}_{\text{total chlorine}}(t) = f[x_{\text{EC}}(t), x_{\text{pH}}(t), x_{\text{temperature}}(t), x_{\text{TOC}}(t), x_{\text{turbidity}}(t), x_{\text{total chlorine}}(t-1)] \quad (7)$$

The left side of p art A of Figure 2 depicts measured versus estimated total chlorine partial training set of the ANN model. Table S1 of the Supporting Information lists the mean, STD, and R^2 statistics for measured and estimated training data of all six water quality parameters. It can be seen that all ANN models have similar means and STD to measured data, and high R^2 (above 0.9), with an exception of turbidity and TOC with R^2 = 0.639 and 0.686, respectively.

2. *Residual Estimation and Classification.* Residuals are calculated for each new observation of each quality parameter as the total error (difference between measured and estimated values) according to eq 3. After observing the behavior of the residuals, the normal residual range of each of the water quality indicators was determined such that majority of the residuals (between 96 and 99%) resides within this range and is acceptable. The left side of Part B of Figure 2 depicts the error between the ANN model and the measured total chlorine of the training data and specified thresholds used to discriminate between outliers and standard errors. As can be seen, the total errors are relatively small and randomly vary around a constant mean, which is to be expected, because the ANN model was trained based on that data. If, at some point, the residuals will exhibit non stationary behavior this will cause a drift in their trend and a gradual classification of all new residuals as outliers. This can be addressed to either the sensors or the ANNs require recalibration. Table S1 of the Supporting Information lists the MSE of the ANN model (training and testing) of all six water quality parameters. The results demonstrate small MSE for all water quality parameters considering their measuring scale.

3. *Model Assessment.* The final stage of the preparation is to evaluate the sensitivity and specificity of the ANN model. For each parameter, TPR and FPR are calculated using eq 5 based on residuals classification and outlier identification from the previous step. The left side of parts C and D of Figure 2 illustrates the simulated and the estimated total chlorine and the consequent errors during simulated event. Part f of Figure 2 demonstrates multiple outliers identified during an event [22 948–23 427] time steps. Although the simulated has very small impact on total chlorine, Figure S3 of the Supporting Information shows that this event also has almost no impact on EC and TOC and a noticeable effect on the rest of the parameters. The ANN model was trained to estimate the interplay between the parameters, hence although chlorine itself does not show great deviation from normal operation, three of the other parameters do, hence the interplay are violated and the event is identified. Table S1 of the Supporting Information

Table 1. Probability Updating

time	Monday 10:00	Monday 10:20	Monday 10:40	Monday 11:00	Monday 11:20		Monday 12:20	Monday 13:20	Monday 14:20	Monday 15:20	Monday 16:20	Monday 17:20	Monday 18:20
true state	false	true	true	true	true	⋮	true	true	true	true	true	false	false
Probability of Event													
total chlorine [mg/L]	4.19×10^{-5}	3.06×10^{-3}	4.05×10^{-1}	**9.50×10^{-1}**	**9.50×10^{-1}**	⋮	8.69×10^{-1}	3.10×10^{-2}	4.18×10^{-2}	**9.50×10^{-1}**	**9.50×10^{-1}**	2.73×10^{-1}	2.95×10^{-3}
EC [mS/cm]	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	⋮	2.59×10^{-4}	**9.50×10^{-1}**	8.01×10^{-1}	3.02×10^{-1}	4.16×10^{-2}	5.36×10^{-3}	6.70×10^{-4}
pH	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	⋮	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}
temperature [°C]	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	⋮	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}
TOC [ppb]	1.87×10^{-5}	9.46×10^{-2}	**9.50×10^{-1}**[a]	**9.50×10^{-1}**	**9.50×10^{-1}**	⋮	**9.50×10^{-1}**	**9.50×10^{-1}**	**9.50×10^{-1}**	**9.50×10^{-1}**	4.02×10^{-1}	3.21×10^{-2}	1.77×10^{-3}
turbidity [NTU]	1.00×10^{-5}	1.00×10^{-5}	1.00×10^{-5}	5.34×10^{-2}	**9.50×10^{-1}**	⋮	**9.50×10^{-1}**	**9.50×10^{-1}**	**9.50×10^{-1}**	**9.50×10^{-1}**	1.57×10^{-1}	4.37×10^{-3}	1.08×10^{-4}
Raised Alarms													
alarm (1)[b]	false	false	**true**	**true**	**true**	⋮	**true**	**true**	**true**	**true**	**true**	false	false
alarm (2)	false	false	false	**true**	**true**	⋮	**true**	**true**	**true**	**true**	false	false	false
alarm (3)	false	false	false	false	**true**	⋮	**true**	**true**	false	**true**	false	false	false
alarm (4)	false	false	false	false	false	⋮	false	false	false	false	false	false	false
alarm (5)	false	false	false	false	false	⋮	false	false	false	false	false	false	false
alarm (6)	false	false	false	false	false	⋮	false	false	false	false	false	false	false

[a] 9.4×10^{-1} – Above threshold probability indicating an event, alarm (1). [b] At least one quality indicator raised an alarm.

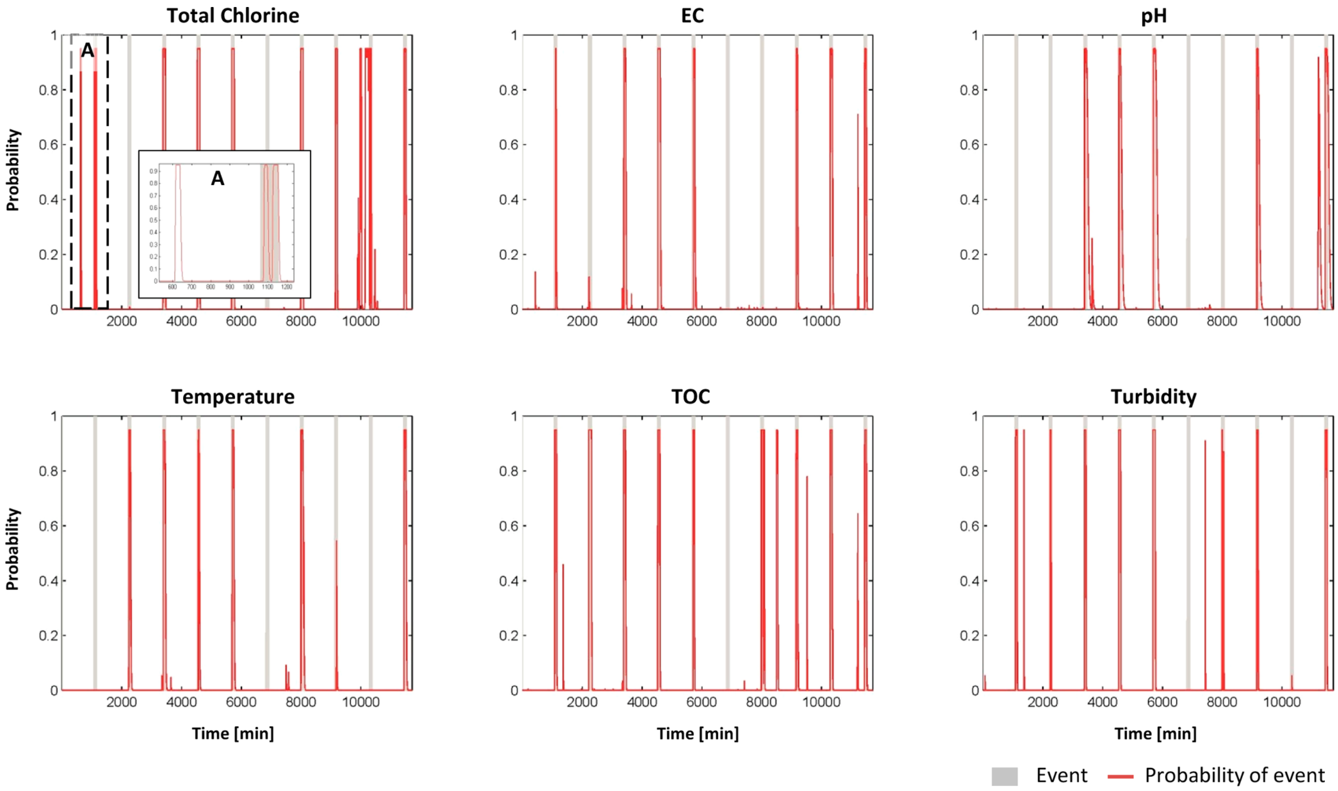

Figure 3. Univariate event probability – six plots demonstrating the likelihoods of a quality event based on Bayesian sequential updating for each quality indicator individually and the true events. (**A**) Example contamination event and estimated event probability ([24 035–24 065] false positive, [24 484–24 579] true positive).

lists the TPR, FPR, and the area under ROC curves for all water parameters that were established during model training. The table shows that TPR is in the range of [0.085–0.587] with total chlorine having the highest TPR and pH – the lowest; FPR is in the range of [0.001–0.093] with temperature having the lowest rates and total chlorine – the highest; and ROC area in the range of [0.549–0.791] with pH and total chlorine having the smaller and the larger areas, respectively.

Model Validation. *1. Parameter Estimation, Residual Evaluation and Classification.* The remaining unseen data (33% of the original data set) was used for testing imitating real-time water quality measurements. For each new observation and for each parameter, the value of the target parameter and the error were estimated using the ANN model created in step 1. The right side of parts A–D of Figure 2 shows the testing of the ANN model and the estimated residuals under regular operation conditions and with contamination events. Figure S3 of the Supporting Information shows the corresponding contamination events for the same time window as in Figure 2. Parts a–d of Figure 2 focus on time period of 24 484–24 579 time steps demonstrating: (a) Measured and estimated total chlorine during normal operation, (b) error between measured and estimated during normal operation, (c) simulated and estimated chloramine during an event, and (d) error between simulated and estimated during an event. From the plots, it can be seen that, during normal operation, model's errors are relatively small, whereas during events the errors are larger and exceed specified thresholds. Parts e–g of Figure 2 show errors classified as outliers during normal operation and part d of Figure 2 – during event, however after further analysis, successive outliers in (d) will evolve into true event, outliers in (g) – into false event, and the error in (e) will be attributed as an outlier. Table S1 of the Supporting Information lists models' and errors' statistics of the testing data. From the table, it can be seen that the mean and STD of the ANN model for all parameters during testing remain relatively close to the measured data; however, R^2 and MSE differ significantly. This can be addressed to the fact that R^2 and MSE are highly susceptible to outliers and noisy data, hence should be replaced with more robust estimators.

2. Event Probability Updating. For each new observation and for each parameter, the probability of the event is updated using sequential Bayesian rule based on eqs 4 and 6 depending on the new observation being classified as an outlier or not and on the TPR and FPR calculated during model preparation. In this application, the initial probability of a contamination event was set to $\pi_1(0) = \pi_0 = 10^{-5}$, the threshold probability for declaring an event was set to $\pi_1(T) = \pi_T = 7 \times 10^{-1}$, and the smoothing parameter was set to $\alpha = 0.6$. At this stage, the probability of an event is updated for each parameter individually and an event is declared based on a single parameter.

Table 1 demonstrates the updating of probabilities for each parameter for a random event. It can be seen how the probabilities gradually increase during an event and decrease when the event ends and the corresponding alarms when the probability exceeds the threshold. The upper bound of the probability was set to 0.95 to prevent it from converging to 1.0, which will keep it from decreasing when the event has passed. Figure 3 graphically illustrates the univariate event probability for each parameter during simulated events. For example, total chlorine detects 8 of 10 events with 3 false alarms, and turbidity – 8 out of 10 events with 2 false alarms. Part A of Figure 3 demonstrates true events versus estimated event probability

Table 2. Detection Table

single indicator	total chlorine [mg/L]	EC [mS/cm]	pH	temperature [°C]	TOC [ppb]	turbidity [NTU]
TP	7	6	5	5	9	8
FP	1	1	1	0	3	2
bmultiple indicator	alarm(1)	alarm(2)	alarm(3)	alarm(4)	alarm(5)	alarm(6)
TP	9	9	7	5	5	4
FP	4	3	0	0	0	0

during [24 035−24 579] time steps, corresponding to parts d−g of Figure 2, with true and false alarms, respectively.

3. Fuse Decision. At each time step, the univariate event probabilities are fused to give the multivariate event probability reflecting the likelihood of an event based on all measured parameters. Figure S4 of the Supporting Information shows six event probability plots. Each subplot illustrates the mutual probability of a random event taking into account one or more equally weighted quality parameters. For example, in Figure S4 of the Supporting Information alarm (1), alarm is raised if at least one parameter signaled an event. In this case, most of the events are detected (i.e., high TPR); however, there are many instances of false alarms (i.e., high FPR). Figure S4 of the Supporting Information alarm (2) shows that an alarm is raised if at least two parameters or more signaled an event. In this case, only one event is undetected with one false alarm. Table 2 summarizes averaged results of multiple runs for 10 contamination events. It can be seen that for single quality indicator, turbidity, TOC, and total chlorine have the highest TPRs. For multiple indicators, the optimum trade-off is attained when at least two indicators raised an alarm.

DISCUSSION AND FUTURE WORK

An event detection algorithm aimed at detecting anomalous behavior of water quality parameters was presented herein. The example application has shown ANNs to be a powerful tool for water quality assessment and classification. The ANN models, capable of depicting the temporal variations in water quality parameters caused by simulated contamination events, combined with the Bayesian updating rule constitute a powerful procedure attaining a comprehensive tool for the decision maker. The proposed technique can be adopted to fit any WDS given a set of tailored parameters (e.g., event initial probability, critical probability, and smoothing parameter). The algorithm was tested for different parameters, time steps, and multiple random contamination scenarios. The ANNs were not sensitive to the frequency of transmitted data (minutes or hours), however all observations need to arrive simultaneously. Initial event probability, initially assumed rare, was lowered to 10^{-10} resulting in a similar outcome (i.e., TPR and FPR). The critical probability, which reflects the decision makers' attitude toward risk, was lowered as well demonstrating that the algorithm is not sensitive to the absolute value but to the order of magnitude relative to the initial probability. The smoothing parameter was also altered $\alpha \in \{0.5, 0.9\}$ influencing the time lag until events are declared. Smaller values result in longer time lags because the updated probability has larger memory; correspondingly, higher values result in smaller time lags, however can result in quick probability convergence and premature declaration of an event. In all instances, the beginning of an event can be inferred from the estimated probability by going back in time after the event was detected.

The outcome of the detection algorithm can be summarized using Table 1, Figure 3, and Figure S4 of the Supporting Information. For each new observation, the algorithm numerically and graphically indicates the probability of a quality fault based on single and multiple measured water quality time-series. Given that an event's strength is not known in advance, the most conservative approach will declare an event if at least one indicator identified an event. In such cases, additional analysis should be performed to decrease the number of false alarms.

Further work needs be conducted to test, generalize, and improve the method's performance. Residual classification should be further investigated by, for example, applying a dynamic threshold optimized to the number of historical observations (time period) and their statistics (mean and STD). Additionally, when the decision about a contamination event is based on multiple quality indicators, weights should be assigned to each indicator relative to its prediction power to provide better results, for example, using the area under ROC curves as possible indication of importance.

Additional research is being conducted to integrate data preparation including filtering, processing, and filling in discontinuous measurements to be included in the general event detection scope and to utilize other data-driven models for quality estimation and other classification techniques to achieve a more robust and reliable event detection method.

ASSOCIATED CONTENT

Supporting Information

Additional evidence on EPA potential contaminants tests, ANN architecture, contamination events, alarm system, and model assessment. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) Kroll, D. J. *Securing Our Water Supply: Protecting a Vulnerable Resource*; PennWell Corporation: Tulsa, OK, 2006.

(2) Stoianov, I.; Nachman, L.; Madden, S.; Tokmouline, T. PIPENET: A Wireless Sensor network for pipeline monitoring. *Proceedings of the ACM/IEEE International Conference on Information Processing in Sensor Networks* **2007**, 264−273.

(3) Ellen, M.; Preis, A.; Iqbal, M.; Srirangarajan, S.; Lim, H. N.; Girod, L.; Whittle, A. J. Real time in-network monitoring to improve operational efficiently. *J. of Amer. W. Wo. Ass.* **2011**, *103* (7), 63−75.

(4) USEPA. WaterSentinel: Online Water Quality Monitoring as an Indicator of Drinking Water Contamination, 2005a, For Official Use Only.

(5) USEPA. WaterSentinel: System Architecture, 2005b, For Official Use Only.

(6) Hall, J.; Zaffiro, A. D.; Marx, R. B.; Kefauver, P. C.; Krishman, E. R.; Haught, R. C.; Herrmann, J. G. On-line water quality parameters as indicators of distribution system contamination. *J. of Am. W. Wo. Ass.* **2007**, *99* (1), 66−77.

(7) Kessler A., Ostfeld A., Sinai G. Detecting accidental contaminations in municipal water networks. *J. of W. Res. Plan. and Man. Div., ASCE*, **1998**, *124*(4), 192-198.

(8) Berry, J. W.; Hart, W. E.; Phillips, C. A.; Uber, J. G.; Watson, J. P. Sensor placement in municipal water networks with temporal integer programming models. *J. of W. Res. Plan. and Man. Div., ASCE* **2006**, *132* (4), 218−224.

(9) Krause, A.; Leskovec, J.; Guestrin, C.; VanBriesen, J.; Faloutsos, C. Efficient sensor placement optimization for securing large water distribution networks. *J. of W. Res. Plan. and Man. Div., ASCE* **2008**, *134* (6), 516−526.

(10) Ostfeld, A.; Salomons, E. Optimal layout of early warning detection stations for water distribution systems security. *J. of W. Res. Plan. and Man. Div., ASCE* **2004**, *130* (5), 377−385.

(11) Preis, A.; Ostfeld, A. Multiobjective contaminant sensor network design for water distribution systems. *J. of W. Res. Plan. and Man. Div., ASCE* **2008**, *134* (4), 366−377.

(12) Xu J., Fischbeck P., Small M. J., VanBriesen J., Casman E. Identifying sets of key nodes for placing sensors in dynamic water distribution networks. *J. of W. Res. Plan. and Man. Div., ASCE*, **2008**, *136*(2), 378-385.

(13) Hart, D. B.; Klise, K. A.; Vugrin, E. D.; McKenna, S. A.; Wilson, M. *CANARY user's manual version 4.3*. EPA/600/R-08/040A. U.S. Environmental Protection Agency, Office of Research and Development, National Homeland Security Research Center, Cincinnati, OH, 2010.

(14) Murray, R.; Haxton, T.; McKenna, S. A.; Hart, D. B.; Klise, K. A., Koch, M.; Vugrin, E. D.; Martin, S.; Wilson, M.; Cruze, V. A.; Cutler, L. Water quality event detection systems for drinking water contamination warning systems: Development testing and application of CANARY. EPA/600/R-10/036, U.S. Environmental Protections Agency, Office of Research and Development, National Homeland Security Research Center, Cincinnati, OH, 2010.

(15) *CANARY, Event Detection Software, EPA*. Website: https://software.sandia.gov/trac/canary

(16) Hart D.; McKenna S. A.; Klise K.; Cruz V.; Wilson M. CANARY: A water quality event detection algorithm development tool. *Proceedings of the World Environmental and Water Resources Congress, ASCE, Reston, VA*, 2007.

(17) *HACH GuardianBlueTM Homeland Security Technologies*. Website: http://hachhst.com

(18) Edthofer F., van den Broeke J., Ettl J., Lettl W., Weingarthner A. Reliable online water quality monitoring as basis for fault tolerant control. *Conf. on Cont. and Fault Tolerant Sys.*, IEEE, Nice, France, 2010, 57-62.

(19) Chungsying, L.; Biswas, P.; Clark, R. M. Simultaneous transport of substrates, disinfectants and microorganisms in water pipes. *Water Res.* **1994**, *29* (3), 881−894.

(20) Rauch, W.; Vanhooren, H.; Vanrolleghem, P. A. A simplified mixed-culture biofilm model. *Water Res.* **1999**, *33* (9), 2148−2162.

(21) Munavalli, G. R.; Mohan Kumar, M. S. Dynamic simulation of multi-component reaction transport in water distribution systems. *Water Res.* **2004a**, *38* (8), 1971−1988.

(22) Milot, J.; Rodriguez, M. J.; Sérodes, J. B. Contribution of neural networks for modeling trihalomethanes occurrence in drinking water. *J. of W. Res. Plan. and Man. Div., ASCE* **2002**, *128* (5), 370−376.

(23) Rodriguez, M. J.; Milot, J.; Sérodes, J. B. Predicting trihalomethane formation in chlorinated waters using multivariate regression and neural networks. *J. of W. Sup.: Res. Tech. − AQUA* **2002**, *52* (3), 199−215.

(24) Rodriguez, M. J.; West, J. R.; Powell, J.; Sérodes, J. B. Application of two approaches to model chlorine residuals in Severn Trent Water LTD (STW) distribution systems. *Water Sci. Technol.* **1997**, *36* (5), 317−324.

(25) Rodriguez, M. J.; Sérodes, J. B. Assessing empirical linear and non-linear modeling of residual chlorine in urban drinking water systems. *Env. Mod. & Soft.* **1999**, *14* (1), 93−102.

(26) Serodes, J. B.; Rodriguez, M. J.; Ponton, A. Chlorcast© a methodology for developing decision-making tools for chlorine disinfection control. *Env. Mod. & Soft.* **2001**, *16* (1), 53−62.

(27) Gibbs, M. S.; Morgan, N.; Maier, H. R.; Dandy, G. C.; Nixon, J. B.; Holmes, M. Investigation into the relationship between chlorine decay and water distribution parameters using data driven methods. *Math. Comp. Mod.* **2006**, *44* (5−6), 485−498.

(28) D'souza, C. D.; Kumar, M. S. M. Prediction of Multi-components (chlorine, biomass and substrate concentrations) in water distribution systems using artificial neural network (ANN) models. *Water Sci. Technol.: W. Sup. − WSTWS* **2009**, *9* (3), 370−376.

(29) Romano M., Kapelan Z., Savic D. A. Pressure signal de-noising for improved real-time leak detection. *Proc. of the 9th. Conf. on Hyd., Tianjin, China*, 2010a.

(30) Romano M., Kapelan Z., Savic D. A. "Real-time leak detection in water distribution systems". *Proc. of the 12th An. W. Dis. Sys. An. Symp., WDSA, Tucson, Arizona, USA*, 2010b

(31) Maeir, H. R.; Dandy, G. C. Neural networks for the prediction and forecasting of water resources variables: a review of modeling issues and applications. *Env. Mod. & Soft.* **2000**, *15* (1), 101−124.

(32) Rumelhart, D. E.; Widrow, B.; Lehr, M. A. The basic ideas in neural networks. *Com. of the ACM* **1994**, *37* (3), 87−92.

(33) Flood, I.; Kartam, N. Neural networks in civil engineering. I. Principles and understanding. *J. of Comp. in Civ. Eng.* **1994**, *8* (2), 131−148.

(34) Hammerstrom, D. Neural networks at work. *IEEE Spectrum* **1993**, 26−32.

(35) McKenna, S. A.; Wilson, M.; Klise, K. A. Detecting changes in water quality data. *J. of Amer. W. Wo. Ass.* **2008**, *100* (1), 74−85.

(36) Rubinstein, R. Y. Kroese, D. P. The cross-entropy method: aunified approach to combinatorial optimization, monte-carlo simulation, and machine learning. Springer-Verlag, 2004.

(37) Klise K. A.; McKenna S. A. Multivariate application for detecting anomalous water quality. *Proc. of the 8th An. W. Dis. Sys. An. Symp.*, WDSA, Cincinnati, Ohio, USA, 2006.